Participation

For Special Participation B, I used ChatGPT 5.1 Standard Thinking on the coding portion of HW 7.

- Here is the link to my chat:
  - https://chatgpt.com/share/69363c35-b9ec-8006-806d-2d5bf040a30f
  - https://chatgpt.com/share/69363dc1-0e20-8006-99be-a5d5ace3b95c
  - Q5: https://chatgpt.com/share/69363e3f-87f0-8006-a054-9b6d8eec7aa5, https://chatgpt.com/share/69363e56-3044-8006-b71f-72208ff5cc80

Formal Code Review (Best for a report or official feedback)

Assessment of ChatGPT 5.1 Standard Thinking Performance on HW7

Executive Summary:
The model demonstrated high proficiency, successfully completing the RNN, Last-Name Classifier, and Autoencoder tasks in alignment with staff specifications. However, the Graph-Clustering module contains two significant mathematical errors regarding the adjacency and degree matrices that require correction to function correctly.

Detailed Findings:

- **Graph-Clustering (Spectral):** This section requires revision.
  - **Critical Defect:** The RBF kernel implementation incorrectly calculated similarity using $exp(+\gamma\|x_i - x_j\|^2)$. This must be corrected to a negative exponent ($-\gamma$) to properly represent similarity.
  - **API Inconsistency:** The function `get_degree_matrix` returns the inverse-square-root ($D^{-1/2}$) rather than the standard Degree matrix (D). While the subsequent Laplacian calculation compensates for this, the naming convention violates the prompt requirements and risks confusion.
  - *Note:* The underlying spectral logic (SVD, row-normalization, KMeans) is otherwise sound.
- **RNN & Gradients:** Excellent implementation. The model correctly structured the `RNNLayer` (two linear layers, single bias, explicit unrolling) and the regression model (shared readout). The gradient visualization tool appropriately targets the recurrent matrix (W_{hh}), making it effective for diagnosing vanishing/exploding gradients.
- **Last-Name Classifier:** A robust, vectorized solution. The pipeline correctly handles padding (gathering the last valid timestep), utilizes plausible hyperparameters (LSTM with dropout, Adam optimizer), and achieves the target accuracy. The response to the ethics prompt regarding deployment was nuanced, highlighting issues like proxy discrimination and privacy.
- **Autoencoders:** Flawless execution. The Vanilla, Denoising, and Masked architectures are correctly implemented with symmetric decoders. The evaluation suite is comprehensive, featuring a linear probe for feature assessment and a plotting helper that accurately renders performance statistics (mean + min/max bands).